THE HOM FUNCTOR OUT OF A DG BIMODULE IS A FUNCTOR OF DG CATEGORIES

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-hom (source lines 5095-5114); the proof environment begins two lines below it and is its own proof, closing at source line 5152.

Editorial note. The statement invokes the author's lemma lemma-functorial by label; that lemma's statement and its own proof are retained as a context span placed before the counted statement, so the reference resolves inside the assembled package and no source pointer is declared.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-dga/preamble.tex, 217 lines), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Differential Graded Algebra chapter, counted once: the Hom functor out of a dg bimodule is a functor of dg categories and induces functors on the module and homotopy categories.

Allowed standard prerequisites (editorial).

- Differential graded category. A category enriched in graded R-modules with a differential on each Hom complex for which composition satisfies the Leibniz rule.. The composition c and its Leibniz rule are used at source lines 5126-5140.
- Dg (A,B)-bimodule and its Hom functor. For a dg (A,B)-bimodule N the assignments on objects and on Hom complexes that make the Hom out of N into an A-module a dg functor.. It is the construction whose compatibility with differentials the proof establishes (source lines 5117-5125, 5131-5134).
- Induced functors on the module and homotopy categories. A functor of dg categories that is compatible with differentials induces functors between the associated module categories and between their homotopy categories (the author's lemma lemma-functorial, retained as context).. It is invoked in the statement itself to pass from the dg functor to the functors on Mod and on K(Mod) (source line 5113).

Lemma 0.1. *Let R be a ring. A functor $F : \mathcal{A} \to \mathcal{B}$ of differential graded categories over R induces functors $Comp(\mathcal{A}) \to Comp(\mathcal{B})$ and $K(\mathcal{A}) \to K(\mathcal{B})$.*

Proof. Omitted. □

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Lemma 0.2. *Let R be a ring. Let (A, d) and (B, d) be differential graded algebras over R. Let N be a differential graded (A, B)-bimodule. The construction above defines a functor*

$$\mathrm{Hom}_{Mod^{dg}_{(B,d)}}(N, -) : Mod^{dg}_{(B,d)} \longrightarrow Mod^{dg}_{(A,d)}$$

of differential graded categories. This functor induces functors

$$Mod_{(B,d)} \to Mod_{(A,d)} \quad \text{and} \quad K(Mod_{(B,d)}) \to K(Mod_{(A,d)})$$

by an application of Lemma 0.1.

Proof. Above we have seen how the construction defines a functor of underlying graded categories. Thus it suffices to show that the construction is compatible with differentials. Let N_1 and N_2 be differential graded B-modules. Write

$$H_{12} = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N_1, N_2), \quad H_1 = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, N_1), \quad H_2 = \mathrm{Hom}_{\mathrm{Mod}^{dg}_{(B,\mathrm{d})}}(N, N_2)$$

Consider the composition

$$c : H_{12} \otimes_R H_1 \longrightarrow H_2$$

in the differential graded category $\mathrm{Mod}^{dg}_{(B,\mathrm{d})}$. Let $f : N_1 \to N_2$ be a B-module homomorphism which is homogeneous of degree n, in other words, $f \in H^n_{12}$. The functor in the lemma sends f to $c_f : H_1 \to H_2$, $g \mapsto c(f, g)$. Similarly for $\mathrm{d}(f)$. On the other hand, the differential on

$$\mathrm{Hom}_{\mathrm{Mod}^{dg}_{(A,\mathrm{d})}}(H_1, H_2)$$

sends c_f to $\mathrm{d}_{H_2} \circ c_f - (-1)^n c_f \circ \mathrm{d}_{H_1}$. As c is a morphism of complexes of R-modules we have $\mathrm{d}c(f, g) = c(\mathrm{d}f, g) + (-1)^n c(f, \mathrm{d}g)$. Hence we see that

$$\begin{aligned}(\mathrm{d}c_f)(g) &= \mathrm{d}c(f, g) - (-1)^n c(f, \mathrm{d}g) \\ &= c(\mathrm{d}f, g) + (-1)^n c(f, \mathrm{d}g) - (-1)^n c(f, \mathrm{d}g) \\ &= c(\mathrm{d}f, g) = c_{\mathrm{d}f}(g)\end{aligned}$$

and the proof is complete. □